

Salifort Motors: Employee Attrition Analysis

The Problem

Salifort Motors faces a high turnover rate of **16.6%**. This analysis aims to identify key drivers of attrition and build a model to predict future resignations.

Summary

Approach	- Cleaned and explored data for 15,000 employees. - Implemented Multiple Logistic Regression as a baseline. - Built and tuned a Random Forest Classifier using GridSearchCV (optimized for F1-score). - Performed feature importance analysis to find key drivers.
Key Insights	Workload is the "Silent Killer": Employees with more than 5 projects or working more than 250 hours/month have a critical risk of leaving. Satisfaction Gap: Low satisfaction scores are the strongest statistical indicator of imminent resignation. Tenure Crisis: A spike in attrition occurs around the 3-5 year mark.
Results	- The final model achieved 98% accuracy and 91% Recall. - The model reliably identifies employees at risk, allowing for proactive retention strategies.
Recom-mendations	Cap Projects: Limit to max 5 per person to avoid certain resignation. Monitor Hours: Flag employees working >240 hours/month. Tenure Focus: Target the 3-5 year group with retention programs.

Model Insights: Key Drivers of Attrition

